Kyle Main

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Data Engineer — Large-Scale Pipelines, Analytics Infrastructure & Abuse-Detection Data Systems

Data engineer with 12 years of experience designing and running the pipelines, warehousing, and analytics infrastructure that power large-scale detection and safety systems — including ETL/ELT ownership for 220+ ingested log/data sources, a company-wide Common Information Model standardizing schema across all of it, and Apache Beam programs run on GCP Dataflow for large-scale historical data retrieval. That foundation sits directly under years of hands-on abuse/threat detection content development and threat-intelligence integration — signature, statistical, behavioral, and ML-based detection logic built on the pipelines and data models I engineered, plus dashboarding and data-quality alerting for stakeholder self-service.

CORE SKILLS

Data Pipelines & ETL/ELT: Pipeline design and ownership for 220+ ingested log/data sources; Common Information Model (CIM) / data dictionary design standardizing field names and types across all ingested data; 50+ Logstash parsing/normalization filters; Apache Beam programs run via GCP Dataflow for large-scale historical/cold-storage data retrieval; connector/collector health monitoring and troubleshooting

Cloud Data Platforms: GCP (BigQuery, Dataproc, Dataflow, serverless event-driven enrichment), AWS, Azure; Elasticsearch as a large-scale analytical data store (Query DSL, transforms, Beats-based ingestion, API)

Distributed Data Processing & Streaming: PySpark / SparkSQL for large-scale exploratory data analysis on GCP Dataproc clusters with Zeppelin notebooks; familiarity with Kafka and Flink for event-streaming workloads

Dashboards, Reporting & Data Quality: Custom Kibana dashboards/visualizations on Elasticsearch transforms for stakeholder self-service reporting; data-quality monitoring and alerting to keep safety-critical data reliable

Abuse & Threat Detection Data Systems: Detection content development (signature, statistical, behavioral, ML-based) on engineered data pipelines — 2,300+ detection rules across the MITRE ATT&CK matrix; threat-intelligence integration into detection logic and alert enrichment

Languages & Data Science: Python, SQL, Pandas, scikit-learn, NumPy; clustering/unsupervised ML; time-series anomaly detection; Git

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint *Oct 2023 – Present*

- Built an entirely new security data ingestion platform for DOE/NNSA from the ground up — CrowdStrike, Suricata, and Zeek telemetry into a central Elasticsearch environment — with data transforms, a UEBA detection layer, data-quality alerting, and custom Kibana dashboards (DOE/NNSA Security Data Integration project, completed).
- Supported data ingestion and data-quality efforts within an Elasticsearch/Splunk environment for DOE's Continuous Diagnostics and Mitigation (CDM) program, a federal continuous-monitoring initiative (CISA CDM at DOE, completed).
- Currently create and manage detection/alerting analytics (Splunk saved searches) directly on top of ingested SOC data supporting Treasury's Security Operations Center incident response and case work (Treasury SOC / TSSOC, current project).

Senior Threat Detection Engineer and Data Scientist — Forescout *Aug 2022 – Oct 2023*

- Senior member of a data science and detection engineering team building signature, behavioral, statistical, and ML-based detection content against massive-scale customer telemetry on a cloud-based big-data platform, incorporating threat-intelligence context from Forescout's in-house research team (Vedere Labs) into detection logic and alert enrichment.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv *Sep 2018 – Aug 2022*

- Owned data engineering for a next-gen cloud SIEM at massive scale as a very early hire — 220+ ingested log sources, 50+ Logstash parsing/normalization filters, and a Common Information Model standardizing field names/types across all of it — plus connector/collector health monitoring and a homegrown Apache Beam program run via GCP Dataflow to fetch large volumes of historical cold-storage data for customers.
- Ran exploratory data analysis at scale on GCP Dataproc clusters using Zeppelin notebooks and a homegrown analysis toolkit — Spark jobs loading data from buckets, PySpark/SparkSQL analysis feeding detection-rule development.
- Created and managed 2,300+ individual detection rules covering most of the MITRE ATT&CK matrix, building out the rules engine and detection content for the startup from scratch, with threat intelligence integrated into rule logic and used directly for false-positive investigation research.

Security Data Scientist — Experian *Jan 2015 – Jan 2018*

- Analyzed large-scale security log data to build custom detection models — DNS-based malware detection/mitigation and anomalous-behavior discovery across the network.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox

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Hiring Team
Anthropic — Safeguards

Robust data infrastructure is the thing that makes safety work possible at scale, and building that infrastructure is what I've spent 12 years doing — most recently underneath detection and abuse-monitoring systems that had to be right, fast, and auditable. I'd like to bring that same foundation to the Safeguards team's data pipelines, warehousing, and analytical tooling.

At Trend Micro/Cysiv, I owned the data engineering for a next-gen cloud SIEM from its earliest days as a startup — ETL/ELT pipelines for 220+ ingested log sources, 50+ Logstash parsing/normalization filters, and a Common Information Model that standardized field names and types across every source so downstream detection and analytics could trust the data. I built a homegrown Apache Beam program run via GCP Dataflow to retrieve large volumes of historical cold-storage data on demand, and ran exploratory data analysis at scale on GCP Dataproc clusters using PySpark and SparkSQL. On top of that foundation, I created and managed 2,300+ detection rules spanning most of the MITRE ATT&CK matrix — signature, statistical, behavioral, and ML-based logic — and integrated threat intelligence directly into that detection content and alert enrichment workflow, the same kind of abuse-pattern-detection work this role's data needs to support.

At Shorepoint, I built DOE/NNSA's security data integration platform from scratch — ingesting CrowdStrike, Suricata, and Zeek telemetry into a new Elasticsearch environment, then layering a UEBA detection model, data-quality monitoring and alerting, and custom Kibana dashboards on top so stakeholders could see model behavior and anomalies without needing to query the raw data themselves. That combination — pipeline ownership, data modeling, dashboarding, and detection content built on a foundation of Python, SQL, and cloud data platforms (BigQuery, Dataproc, Dataflow) — is exactly the range this Safeguards role asks for.

I'd welcome the chance to talk through how that background applies to giving Anthropic's Safeguards team the data foundation it needs to detect misuse and keep model behavior safe at scale.

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